Where Did the Heat Go?

Keeping the system and the surroundings on separate lines of the ledger, and turning a ΔH in kJ/mol into a temperature you could measure with a thermometer.

Name ____________________ Date ____________________ Period ____________________

Demonstration: andresdbp.github.io/resources/demos/where-did-the-heat-go/

1 Before you open anything

Answer from what you already know. Being right is not the point — having a number on paper to argue with later is.

PREDICTION

Two beakers each hold 100.0 g of water at 22.0 °C. In beaker **A**, 0.100 mol of ammonium nitrate dissolves; in beaker **B**, 0.050 mol of the same salt dissolves. For NH_4NO_3, ΔH_{soln} = +25.7 kJ/mol. Predict the final water temperature in each beaker.

Beaker A: _____________ °C   Beaker B: _____________ °C

Write one or two sentences saying how you got those numbers — including what you did with the plus sign.

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2 Reading the four salts

Open the demonstration. The beaker on the left is the surroundings — 100.0 g of water starting at 22.0 °C. The small vessel inside it is the system, the reaction itself. The *Energy accounting* panel on the right shows the arithmetic line by line.

1. Check that **Amount reacting** reads **0.100 mol**. Leave it there for the whole of this part.
2. Under **Real examples**, click **NH_4NO_3**.
3. Fill in that row of the table. The last column comes from the line labelled **final water temp** in the *Energy accounting* panel.
4. Repeat for the other three preset buttons.

Preset	ΔH of the reaction (kJ/mol)	System — energy (kJ)	Surroundings ΔT (°C)	final water temp (°C)	Hand feels?
NH_4NO_3					
NH_4Cl					
NaOH					
$CaCl_2$					

a. In every row, compare the sign of *ΔH of the reaction* with the sign of *Surroundings ΔT*. State the relationship in one sentence.

b. Look at the boxes marked *q system* and *q surroundings*. Write down what the box marked *system + surroundings* reads for each of the four presets, then say what physical law that box is displaying.

c. For NaOH the demonstration reports ΔH = -44.5 kJ/mol and yet the water gets hotter. Explain in one or two sentences why a negative number and a temperature rise are not in conflict here.

3 How much of it reacts

Now hold the chemistry fixed and change only the quantity. Click **NH_4NO_3** so that ΔH stays at +25.7 kJ/mol, then work down the **Amount reacting** slider. It moves in steps of 0.005 mol; click it once and use the arrow keys if dragging is imprecise.

Amount reacting (mol)	**System — energy (kJ)**	**Surroundings ΔT (°C)**	**final water temp (°C)**	**ΔT ÷ amount (°C/mol)**
0.025				
0.050				
0.075				
0.100				

a. State the rule you have just measured, in your own words: what happens to *Surroundings ΔT* when the amount reacting is doubled, and what stays constant down the last column?

b. The last column is constant but the *final water temp* column is not proportional to the amount. Explain why those two statements are both true.

c. Go back to your prediction in Part 1. Compare it with the 0.100 mol and 0.050 mol rows. If it was off, name the specific assumption that turned out not to hold.

4 Working backwards, and a shortcut that fails

A test question rarely hands you ΔH and asks for ΔT. More often it hands you the thermometer reading and asks for ΔH.

1. Set **Amount reacting** to **0.100 mol** and select the **Endothermic (+ΔH)** tab.
2. On paper, work out the ΔH that would take the 100.0 g of water from 22.0 °C down to exactly **12.0 °C**. Use $q = mc\Delta T$ with c = 4.18 J g^{-1} $°C^{-1}$, then convert to a per-mole quantity.
3. Set the **ΔH of the reaction** slider to your answer (it moves in steps of 0.1 kJ/mol) and check *Surroundings ΔT* and *final water temp*.

YOUR WORKING

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ΔH from your working: _____________ kJ/mol *Surroundings ΔT* shown: _____________ °C *final water temp* shown: _____________ °C

A reasonable shortcut says: the bigger the enthalpy of solution, the bigger the temperature change. Test it.

1. Click **$CaCl_2$** (ΔH = -82.8 kJ/mol) and set **Amount reacting** to **0.020 mol**.
2. Click **NaOH** (ΔH = -44.5 kJ/mol) and set **Amount reacting** to **0.100 mol**.

Set-up	ΔH of the reaction (kJ/mol)	System — energy (kJ)	Surroundings ΔT (°C)
$CaCl_2$, 0.020 mol			
NaOH, 0.100 mol			

a. Which set-up has the larger ΔH per mole, and which produces the larger temperature change? Say precisely what the shortcut leaves out.

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The caption under the beaker states the simplification: every joule is assumed to pass between the reaction and the 100.0 g of water, with nothing going to the glass, the thermometer or the air. A real coffee-cup calorimeter also warms its own container, so a measured ΔT comes out somewhat smaller than the one on screen. Any ΔH you calculate from that measurement without correcting for the calorimeter is therefore an underestimate of the true magnitude.

5 Solve these without the demonstration

Close the page, or look away from it. Show your working. Then reopen it, set up each situation, and check yourself — and where you were wrong, write down which step went off. Take 100.0 g of water starting at 22.0 °C, with c = 4.18 J g^{-1} $°C^{-1}$, unless told otherwise.

1. 0.060 mol of NH_4Cl dissolves in the water; ΔH_{soln} = +14.8 kJ/mol. Find q_{system}, $q_{surroundings}$, ΔT, and the final water temperature. Label each sign and say in a few words what each sign means physically.

2. A hot pack must bring the 100.0 g of water from 22.0 °C to 35.0 °C using $CaCl_2$, ΔH_{soln} = -82.8 kJ/mol. How many moles must dissolve? The demonstration's amount slider moves in steps of 0.005 mol — state the nearest value it can take and the temperature that value would actually reach.

3. Which gives the larger rise in water temperature: 0.030 mol of a salt with ΔH_{soln} = -82.8 kJ/mol, or 0.060 mol of one with ΔH_{soln} = -44.5 kJ/mol? Decide by calculation, not by inspection, and give both values of ΔT.

4. A lab report states: "ΔH = -44.5 kJ/mol, so the reaction released negative energy and the reaction got hot." Two things in that sentence are wrong. Identify both, and rewrite the sentence correctly using the words *system* and *surroundings*.

6 On your own

Nothing here is graded. The demonstration does more than this worksheet uses.

- Set the **Endothermic (+ΔH)** tab, 0.100 mol, and push the ΔH slider to its maximum of 85.0 kJ/mol. An orange caution note appears about the bath approaching freezing. Find the smallest ΔH at which it appears, then work out from $q = mc\Delta T$ why the model stops being trustworthy there.
- Turn the amount down to 0.020 mol with NH_4Cl selected. The note under the beaker changes to say the change is too small to feel. Raise ΔH until it changes back, and record roughly what ΔT the boundary sits at.
- Flip between the **Endothermic** and **Exothermic** tabs without touching either slider, watching the enthalpy diagram. List what changes and what does not — the height of the arrow, its direction, the relative positions of reactants and products.
- Watch the two bars in the ledger at the bottom of the beaker panel while a run is in progress, using **Replay**. They are mirror images at every instant, not just at the end. Say what that means about when energy conservation applies.